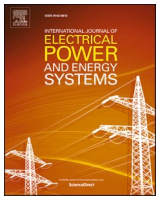
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Security assessment method for inertia and frequency stability of high proportional renewable energy system

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ABSTRACT

With the increase in large-scale renewable energy and DC transmission, synchronous generators in power grids are gradually being replaced, resulting in a continuous reduction in inertia. The resulting low-inertia system influenced the frequency stability of the power grid. On the one hand, knowledge of the inertia level of the system is relevant to evaluate the frequency stability of the power system. On the other hand, knowledge of the anti-interference ability of the system is also relevant. Based on the frequency response mechanism of a conventional synchronous system, this study extended the concept of inertia of a synchronous generator by introducing virtual inertia and proposed an inertia evaluation method for power grids with high-penetration electronic devices. Subsequently, a minimum inertia estimation method considering fast frequency response was proposed based on the virtual inertia technology of grid-connected inverters. Finally, the effectiveness of the proposed model was illustrated using a case study based on the South-East Australian system.

1. Introduction

Owing to the threat of the depletion of global fossil energy and the increasingly severe greenhouse effect, an upsurge in clean energy is being established worldwide and the energy structure dominated by fossil energy is gradually shifting to one dominated by wind energy, photovoltaics, hydropower, and other renewable energy sources [1]. Many countries and regions have proposed the idea of building 100% renewable energy power systems. More than 60 countries had proposed 100% renewable energy power system goals by the end of 2019 [2]. When a conventional power grid dominated by synchronous units gradually transitions to a power system with a high proportion of asynchronous resources, the inertia and primary frequency modulation capacity provided by synchronous units decrease continually and the anti-disturbance ability of the system decreases [3]. For example, the "9.28" power outage of the South Australian power grid in 2016 and the "8.9" power outage in Britain in 2019 [4,5]. When a power disturbance occurs in a power system dominated by a synchronous machine, power support is provided by the synchronous inertia response, primary frequency modulation, and secondary frequency modulation [6].

The installed capacity of renewable energy sources will continue to rapidly expand in the future. Regarding the system frequency stability [7], the system inertia under different renewable energy penetrations cannot be evaluated, restricting further improvement in renewable energy consumption. Therefore, estimating the inertia of large-scale systems under different renewable energy output ratios is crucial to determine the frequency support capacity of the system after a disturbance, guiding the accurate operation of the power grid, and accelerating the construction of power systems with renewable energy [8]. The inertia is crucial in maintaining frequency stability. Therefore, a logical evaluation of the inertia requirement of a system is the basic problem for the frequency stability of a high proportion of renewable energy power systems [9]. Recently, some researchers introduced the frequency stability constraint into the unit commitment model of power generation dispatching [10–12], obtained an economic operation scheme to ensure the frequency stability of the system, and investigated the online inertia evaluation method of system operation based on synchronous phasor measurements [13]. An inertia emulation method that uses the frequency of the center of inertia to slow the frequency change rate and increase the frequency nadir during frequency-dropping events was proposed [14]. A novel application of virtual inertia emulation to emulate inertia power, which can improve frequency performance and stability, was presented in [15]. Furthermore, [16] proposed a strategy using the energy stored in a supercapacitor connected to a back-to-back

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converter DC link to emulate the inertial response. However, they determined the minimum inertia required by the system through the constraints of the rate of change of frequency (*RoCoF*) and frequency nadir. Therefore, the system inertia of different renewable energy outputs should be considered to determine the frequency support capacity of the system after a disturbance, which can realize the foresight of inertia evaluation.

However, the inertia response has a delayed effect on the frequency stability. It may not wait for the primary frequency regulation and the system triggers a protective action. Therefore, some countries and regions faced with the problem of low inertia have designed a fast frequency response (FFR) [17] to resolve the shortage of synchronous machine inertia and primary frequency modulation reserve capacity, as well as limit the *RoCoF* before the primary frequency regulation response through active power fast injection. Owing to the fast response characteristics of FFR, its action period partially coincides with the inertia response. Fradley proposed the use of an adaptive frequency control implementation to provide FFR within low-inertia systems as power electronic interfaced energy sources [18]. In [19], capitalizing on the rapid action characteristics of renewable energy, the authors proposed a method that considers FFR to improve the participation of renewable energy. Additionally, [20] proposed a frequency response estimation based on a machine learning algorithm, which could improve power system stability. However, they did not analyze different virtual inertia (VI) technologies such as grid-forming and grid-following VI. Therefore, the controllable elements in the system should be fully utilized—mainly the large-scale access to renewable energy—to participate in the FFR of the system through inverter control.

Currently, directly estimating the inertia under the proportion of renewable energy output is rare. Generally, related studies focus on the evaluation of the system inertia and estimation of the minimum inertia. In converter-dominated power systems, many synchronous generators are replaced by converter-interfaced generations, which are initially inertia-free because of their operational characteristics. An expression for the equivalent inertia was derived, the inertia response of the power system was analyzed, and the equivalent inertia was estimated using system identification [24]. The evaluation of the inertia and primary frequency regulation capability of a power source node using a non-invasive online estimation method from the impedance perspective was investigated in [25]. An approach to estimate the aggregated inertia at the point of interconnection, which can be further used to calculate the total inertia of the power system and to monitor the inertia spatial distribution was proposed [26]. Most current methods estimate the inertia and minimum inertia under the real-time output proportion using the measurement data of frequency or power change, emphasizing real time. However, in a large-scale power system, owing to a large number of synchronous machines and different inertia constants, the start and stop status change with the load and other factors, including virtual inertia. Thus, determining the frequency in the center of inertia is challenging, vulnerable to signal interference, real-time calculation results cannot be obtained, and an accurate mastery of the system inertia under the current renewable energy output is challenging.

As shown in Fig. 1, in a low-inertia power system, the inertia response and frequency control are crucial and the FFR process of the new energy is equivalent to the inertia and frequency response process of a synchronous generator. This study describes the source of inertia in a power system after a high penetration of renewable energy, a calculation method for the upper and lower bounds of the inertia estimation of the power system, as well as the solution of the minimum inertia requirement index, considering the FFR of power electronics. Subsequently, the effectiveness of the method was validated through a simulation, which provides a reference for the frequency stability analysis of the new power system.

The contributions of this study are as follows: 1) Different unit startup modes and inertia levels will exist for the same renewable energy output ratio. No linear relationship exists between the output ratio and the system inertia. The inertia should be estimated under a particular renewable energy output ratio. We theoretically explored the internal relationship between renewable energy output and inertia. To address the problem of the unknown scale and output proportion of the power system with renewable energy, data measurement and real-time estimation of inertia were conducted to provide auxiliary decision-making for dispatchers. 2) According to typical measurement data and the preset startup state, the inertia level of the system after the renewable energy output is increased was estimated, which can be applied to formulate the power grid operation mode in combination with the minimum inertia of the system. The system inertia range under this renewable energy output ratio can be estimated using the renewable energy output ratio to infer the starting capacity range of the synchronous machine, combined with the pre-start renewable energy unit configuration with virtual inertia. 3) Combined with the correlation between inertia and frequency and considering the fast frequency response, the maximum disturbance that can be withstood by the system under a certain renewable energy output proportion can be directly obtained. The fluctuation range of the system frequency under a certain disturbance can be determined to guide the adjustment of the renewable energy output proportion in the system, reasonable and orderly arrangement, building renewable energy units with virtual inertia, and improving the overall control of the power system.

The remainder of this paper is organized as follows: In Section II, the definitions of inertia and FFR are introduced and the inertia estimation considering the penetration of renewable energy and minimum inertia requirement considering FFR are presented in Section III. In Section IV, a case study is conducted on the modified 14-machine 59-bus system located in South-East Australia to validate the effectiveness of the method. Finally, Section V concludes the paper.

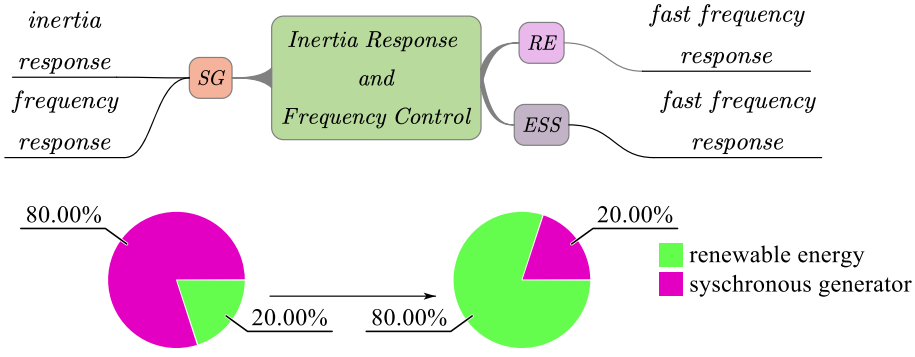


Fig. 1. Low-inertia power system.

2. System model

2.1. Definition of inertia

The inertia of the power system can resist the frequency change caused by external disturbances and slow the *RoCoF*, which guarantees the system frequency stability. The swing equation of the *i*-th synchronous generator can be formulated as follows:

$$2\frac{H_i}{f}\frac{d\Delta w_i}{dt} = \Delta P_{m,i} - \Delta P_{e,i} - D\Delta w_i \tag{1}$$

$$H_i = \frac{E_i}{S_{B,i}} \tag{2}$$

where H_i is the inertia constant of the *i*-th synchronous generator, Δw_i is the deviation of the rotor speed, $\frac{d\Delta w_i}{dt}$ is the *RoCoF*, $\Delta P_{m,i}$ and $\Delta P_{e,i}$ are the mechanical and electrical powers, respectively, and f is the rated frequency. Furthermore, D is the damping factor, $S_{B,i}$ is the rated power, and E_i is the kinetic energy of the synchronous machine, which can immediately realize spontaneous inertial support.

However, an increasing number of conventional synchronous generators are being replaced by power electronics interfaced with renewable generation. With an increase in the penetration of renewable energy and a reduction in the number of synchronous generators, the system inertia level and primary frequency regulation capacity are reduced simultaneously. The inertia provided by the synchronous machine alone does not satisfy the minimum inertia requirement of the system. It must be provided jointly by the inertia of the synchronous generators and the virtual inertia from power electronics.

In the case of high penetration of renewable energy, the virtual inertia of power electronics is the main component of system inertia. However, owing to the short delay in the grid-following virtual inertia, the instantaneous frequency of the disturbance may change rapidly, threatening the safety of the power grid and damaging the equipment. Moreover, a certain proportion of power electronics should be controlled as a voltage source, that is, grid-forming virtual inertia, considerably to provide immediate inertia support. Therefore, considering the inertial support capacity of the new energy, the swing equation of the power system is expressed as follows:

$$H_{sys} = \frac{E_{SG,total} + E_{RE,total}}{S_{SG,total} + S_{RE,total}} \tag{4}$$

$$E_{SG,total} = \sum_{i=1}^{n_{SG}} H_i S_i \tag{5}$$

$$E_{RE,total} = \sum_{j=1}^{n_{RE}} H_j S_j \tag{6}$$

$$2\frac{H_{sys}}{f}\frac{d\Delta w}{dt} = \Delta P_m - \Delta P_e - D\Delta w \tag{7}$$

where $E_{SG,total}$ and $E_{RE,total}$ are the total kinetic energies of the synchronous generator and renewable energy, respectively. H_{sys} is the rotational inertia constant of the regional system.

Virtual inertia devices can exhibit the same performance as synchronous generators in terms of inertial response. Two virtual inertia devices exist: grid-following and grid-forming virtual inertia devices [21]. The design of grid-following virtual inertia devices is simple and uses a frequency estimator—a phase-locked loop—to obtain the frequency and *RoCoF*. However, this requires a short delay (approximately 100 ms). Grid-forming virtual devices can provide no delay inertia support; however, they are complex and expensive. The source of virtual inertia can be derived from the rotating kinetic energy of renewable energy or the energy storage system. Owing to the uncertainty of renewable energy, the rotating kinetic energy cannot provide stable inertia support, whereas virtual inertia realized by energy storage can significantly increase the equivalent inertia level of the system. Therefore, in a power system with low renewable energy penetration, to prevent the rapid change of frequency caused by instantaneous disturbance, the minimum inertia can comprise instantaneous synchronous and grid-following virtual inertia to simultaneously ensure the inertia requirement and reduce the investment. However, the short delay of the grid-following virtual inertia device may result in rapid frequency change at the instance of disturbance, which threatens the safety of the power grid and damages the equipment. Therefore, with improvements in renewable energy penetration, grid-forming virtual inertial devices have more application prospects. The interconnection of grid-forming virtual inertia devices is as follows [22]:

$$\dot{\theta}_{VI} = w_{VI} \tag{8}$$

$$2\frac{H_{vi}}{f}\frac{dw_{VI}}{dt} = -D_{VI}w_{VI} + P_{VI} \tag{9}$$

where x_s is the state vector containing (but not limited to) the mechanical states of the generators, their controllers, and the states of other devices (that is, nonlinear dynamic loads and renewable in-feed). z_s comprises the AC signals of the transmission network, such as the transmission line currents and bus voltages. The dynamic model of the power system is shown in the green oval diagram in Fig. 2. θ_{VI} and w_{VI} are the angle and frequency of the voltage (that is, $\theta_{VI} = \angle v_{VI}$) generated by the grid-forming virtual inertia device, respectively, and P_{VI} is the active power generated by the virtual inertia device. H_{VI} and D_{VI} are the inertia and damping constants of the virtual inertial device, respectively.

P_{max} is the rated power of the converter; w_{max} and $|\dot{w}|_{max}$ are the maximum deviations of the frequency and *RoCoF*, respectively. Equation (10) represents the constraint on the primary control reserve markets. Furthermore, the inertia and damping responses must also satisfy constraints (11) and (12).

$$\sum_{i=1}^{n_{RE}} D_{VI,i} \le D_{sum} \tag{10}$$

$$H_{VI} \le \frac{P_{max}}{|\dot{w}|_{max}} \tag{11}$$

$$D_{VI} \le \frac{P_{max}}{w_{max}} \tag{12}$$

where D_{sum} is an upper bound on the sum of damping gains.

2.2. FFR

Because converter-interfaced generation operates at a considerably faster time scale compared with the conventional synchronous generation, we treat virtual inertia and fast frequency response as remedies for the stability of low-inertia power systems. The control parameters of the fast frequency regulation of renewable energy based on inverter control

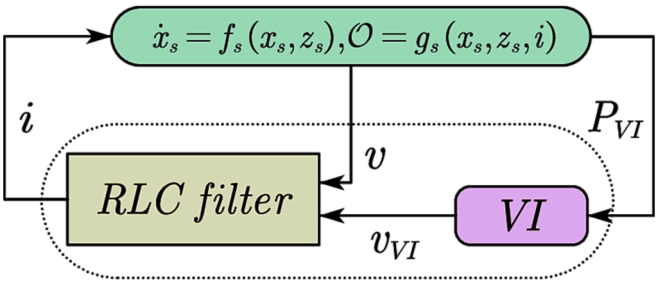


Fig. 2. Interconnection of grid-forming virtual inertia device.

are not subject to physical constraints and can be designed according to system requirements.

The *RoCoF* represents the rate of system frequency change and explains the relationship between the system frequency change as well as the scale of disturbance and inertia of the system. *RoCoF* is generally calculated as an estimation on a particular time window because it cannot be accurately determined in waveform measurement on a short time scale, and the time window is defined as 100–500 ms [23]. After deforming Equation (7), the following equation can be obtained:

$$\frac{d\Delta w}{dt} = \frac{\Delta P^* f}{2H_{sys}} \tag{13}$$

The frequency response curve considering FFR is shown in Fig. 3. Owing to the rapidity of FFR, it partially coincides with the inertial response. In a low-inertia power system, numerous FFRs are required to provide auxiliary frequency-regulation services. Frequency regulation without considering the system inertia level is unsafe and the inertia requirement without considering the system frequency modulation ability is uneconomical.

In a synchronous machine-dominated power system, when a large power disturbance occurs in the system, power support is provided by the synchronous inertial response, primary frequency regulation, and secondary frequency regulation. The synchronous inertia plays a role at approximately 0–3 s after the disturbance. Influenced by the inherent dead zone and response speed of the unit, the primary frequency regulation resource can respond after 2 ~ 30 s. The *RoCoF* in a low-inertia power grid is generally very high; thus, before the primary frequency regulation response, the frequency drop caused by the disturbance is likely to exceed the threshold of the low-frequency load shedding device. Therefore, as a new product, FFR has the rapidity of the inertia response process and the active support capability of primary frequency regulation, that is, the shaded part in Fig. 3.

The system frequency response model is shown in Fig. 4, which considers the inertia response of synchronous generators and the FFR of virtual inertia devices. H_{SG} and H_{VI} are the inertia constants of the synchronous generators and virtual inertia devices, respectively. D_{SG} and D_{VI} are the damping constants of the synchronous generators and virtual inertia devices, respectively.

Moreover, improving the inertia monitoring ability can support the formulation of more accurate inertia requirement assessment indicators. Additionally, it can guide the stable operation of power systems. In real-time operations, owing to the coupling relationship between the inertia response and frequency regulation, clarifying the inertia level of the power system has guiding significance for the optimal scheduling and determination of reserve requirements. The inertia level of the power system varies with time. Particularly, in systems with high renewable energy penetration, renewable energy and energy storage do not have the same definite inertia as the conventional synchronous machine; however, they depend on the climatic conditions or state of charge, which are uncertain.

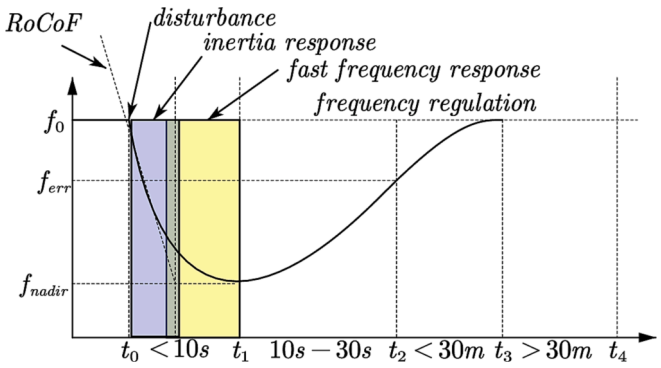


Fig. 3. Frequency response curve considering FFR.

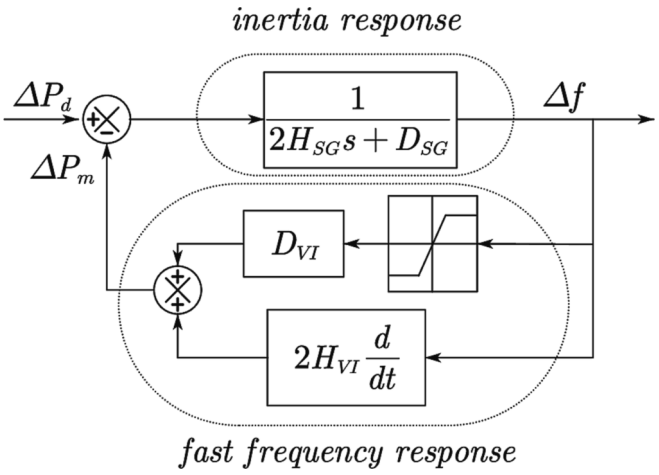


Fig. 4. System frequency response model.

3. Minimum inertia estimation

With continuous access to renewable energy, the inertia of power systems has decreased continually. When the system is subjected to large disturbances, it increases the *RoCoF* of the power system. However, the system may be unstable before the action of the primary frequency regulation and the equipment in the system may be damaged. Therefore, *RoCoF* is crucial for measuring the inertia level of a system. Power system dispatchers require more inertia levels and the minimum inertia requirements of the system. The minimum inertia requirement is expressed as follows:

$$H_{min} = \frac{\Delta P^* f}{2RoCoF_{max}} \tag{14}$$

where H_{min} is the minimum inertia requirement,ΔP is the disturbance that corresponds to changes in the load or generation, and $RoCoF_{max}$ is the maximum *RoCoF*, which can avoid the problem of frequency stability. When the governor does not participate in frequency regulation, the system minimum inertia evaluation based on the frequency deviation constraint can be transformed into a minimum inertia evaluation based on the *RoCoF* constraint.

3.1. Inertia estimation considering renewable energy penetration

The total inertia of the power system can be obtained when the inertia of the system and the virtual inertia are known. The inertia of the synchronous machine is only related to the inertia constant H, rated capacity S_B, and the start and stop status of each unit. The inertia constant and rated capacity of the synchronous generator are non-variables; therefore, the inertia of the synchronous generator is only determined by the startup state of the synchronous generator. Currently, renewable energy is primarily based on conventional methods. Its inertial response is weak under conventional control; thus, its inertia can be ignored. The original calculation result of H_{sys} is no longer applicable when the synchronous generator is gradually replaced by renewable energy. The proportion of replaced synchronous generators and renewable energy that can provide virtual inertia must be analyzed.

$$\alpha = 1 - \varphi + \sigma \tag{15}$$

where α is the proportion of resources that can provide inertia, φ is the proportion of synchronous generators replaced, and σ is the proportion of renewable energy sources that can provide inertia. According to Equation (4), the equivalent inertia H_e of the power system is evaluated as follows:

$$H_e = \alpha^* H_{sys} \tag{16}$$

The installed capacity of synchronous generators and renewable energy in the power system does not change in a short time; thus, only the proportion of synchronous generators and renewable energy should be known in advance to calculate the inertia level of the power system during actual operation.

3.2. Minimum inertia requirement considering FFR

In a conventional power system, only the synchronous generator provides inertia; however, with the synchronous generator gradually replaced by power electronic equipment, the total inertia level of the system decreases. Conversely, the inertia of the power system is time-varying owing to the introduction of virtual inertia. Therefore, the upper and lower bounds of the minimum inertia requirement must be predicted to ensure the frequency stability of the power system.

(1) Upper bound estimation of minimum inertia requirement

Conventional synchronous generators are replaced by converter-interfaced generators; not all converters exhibit virtual inertia. Suppose the proportion of synchronous machines in the power system decreases gradually. In that case, the inertia level of the system is significantly reduced to absorb more new energy and operate with the minimum technical output. The upper bound H_{upper} is as follows:

$$H_{upper} = \frac{\Delta P^* f}{2RoCoF_{max}} \tag{17}$$

The inertia of this condition can be formulated as follows:

$$H_{e1} = H_e \tag{18}$$

(2) Lower bound estimation of the minimum inertia requirement

The FFR mimics the characteristics of the generator by decreasing the response time of power electronics. It can quickly respond to changes in frequency and output active power $P_{VI} = 2H_{VI}\frac{dw_{VI}}{dt} + D_{VI}w_{VI}$ to the power system, which can resist disturbances.

Synchronous generators operate at the rated power; thus, their utilization can be maximized. However, this results in a shortage in renewable energy consumption and all virtual inertia are put into operation. The lower bound H_{lower} is expressed as follows:

$$H_{lower} = \frac{(\Delta P - P_{VI})^* f}{2RoCoF_{max}} \tag{19}$$

The inertia of this condition can be formulated as follows:

$$H_{e2} = H_e / k \tag{20}$$

where k is the percentage of the minimum technical output. Owing to the different geographic locations, generator capacities, and types, the minimum technical output percentage of synchronous machines differs. The average value of the synchronous units in the system can be obtained from statistical data. The average minimum technical output percentage of synchronous machines is defined as the ratio of the sum of the minimum output percentage and the rated capacity product of all synchronous units in the system to the rated capacity of the system.

For systems with rapid scale expansion or under planning, accurate estimates of inertia cannot be obtained owing to the absence of a typical case for reference. However, because the inertia and virtual inertia of the synchronous machine can be regarded as determined only by the startup status of the unit, the startup capacity range of the synchronous machine can be inferred from the new energy output ratio. Combined with the energy stored by the pre-startup new energy unit with virtual inertia and key parameters obtained by the statistical algorithm, the system inertia range under different new energy output ratios can be estimated.

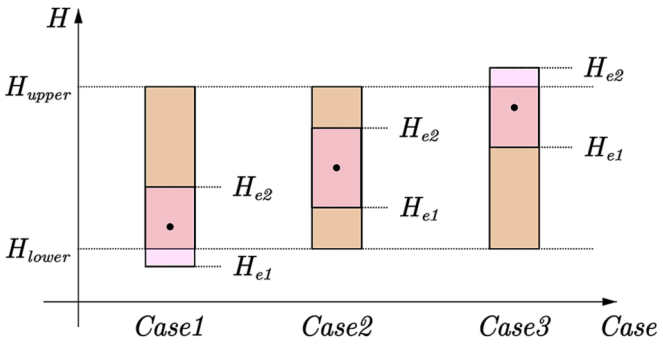


Fig. 5. Diagram of upper and lower bounds.

Three cases were considered, as shown in Fig. 5. The orange box in the figure represents the maximum and minimum inertia requirements of the system. The pink box represents the maximum and minimum inertia estimates for different cases. The inertia demand depends on the power deviation and maximum *RoCoF*, whereas the inertia estimation depends on the proportion of renewable energy and the result of the unit commitment.

In Case 1, some operating points are smaller than H_{lower}, indicating that they are unstable. In Case 2, some operating points are unstable; however, sufficient control measures exist to deal with this case. In Case 3, some operating points are greater than H_{upper}, indicating that they are stable. The dispatcher can better determine the frequency stability of the power system and conduct preventive corrections by obtaining the minimum inertia requirement of the power system in advance.

The algorithm is as follows:

4. Case study

4.1. Case description

A case study was conducted on a modified 14-machine 59-bus system

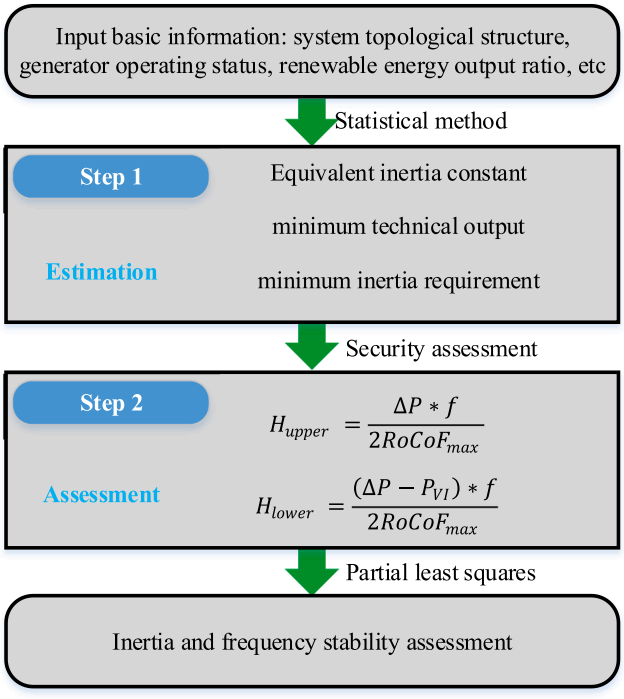


Fig. 6. The framework and process of the method.

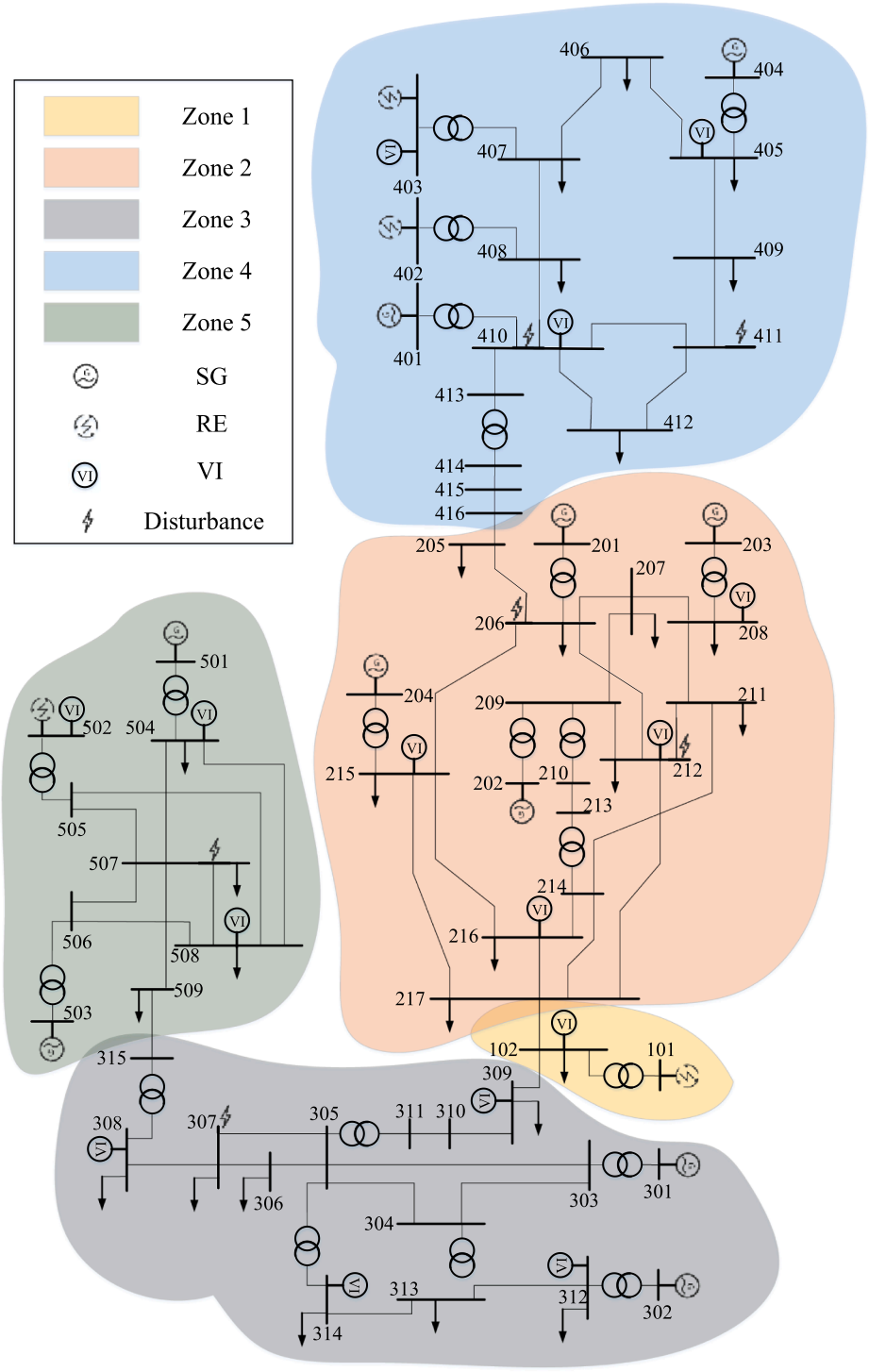


Fig. 7. Modified 14-machine 59-bus system.

located in southeast Australia for illustration [22].

As shown in Fig. 7, this system has a string topology and weak coupling between Area 5 and other areas. To simulate the effect of renewable energy integration, the generators at nodes 101, 402, 403, and 502 were replaced by constant power sources, which could not provide sufficient support. Fifteen VI devices were connected to the grid, as shown in the figure. Additionally, power system stabilizers, governors, and turbines, which are higher-order models, were included in the system. Six disturbances occurred at nodes 206, 212, 307, 411, 502, and 507, which simulated the load increase by changing the active power

Table 1
Generator Parameters.

node	Power (MW)	Inertia (s)	node	Power (MW)	Inertia (s)
201	2000	3.2	302	800	3.5
202	1000	2.8	401	1300	2.6
203	1000	2.6	404	1000	4
204	2000	3.2	501	600	3.5
301	3000	2.8	503	200	7.5
Total power (MW)			12,900		

Table 2
Parameters of constant power sources.

node	Power (MW)
101	800
402	900
403	1300
502	700
Total power (MW)	3700

injection.

The parameters of the generators and constant power sources are listed in Tables 1 and 2 respectively.

In this system, four scenarios were set considering different degrees of penetrations and the local load was maintained at 14807 MW.

4.2. Inertia estimation

Without considering the inertial response of renewable energy, the inertia of this system under different degrees of penetration was calculated according to the data in Table 1, based on Equation (4). The details are as follows.

As shown in Table 3, with the continuous increase in renewable energy penetration and the lack of virtual inertia, the kinetic energy of the system gradually decreases as well as the H_{sys}.

The percentage of minimum technical output *is* $k = 0.7$; the upper and lower bounds of the inertia estimation are as follows:

As shown in Table 4, without considering virtual inertia, the inertia level of the power system decreases with an increase in the penetration of renewable energy. Suppose the generators adopt minimum technical output. In that case, the inertia level in the power system increases, which is uneconomical. The lower bound is more relevant to the dispatcher.

Owing to the continuous increase in the degree of penetration, the inertia of the system gradually decreases, thus decreasing its ability to resist frequency changes, and frequency stability is a problem. The system frequency of different degrees of renewable energy penetration under a disturbance of 200 MW is shown in Fig. 8.

When the proportion of renewable energy increased by 5%, the maximum frequency deviation changed regularly under a disturbance with the same load power growth. The maximum frequency deviation of the system gradually increased with a decrease in inertia.

In Table 5, where $f1$ and $f2$arethe frequencies before and after disturbance, respectively.

Based on the analysis, the frequency decreased significantly with an increase in disturbance. When the inertia of the system is insufficient to resist the frequency change, the threshold of under-frequency load shedding can be easily triggered, that is, 49.5 Hz.

4.3. Minimum inertia requirement

When the inertia level of the power system is predicted in advance, the next step is to predict the minimum inertia requirement to maintain the frequency stability of the power system; that is, the sum of the synchronous machine inertia and virtual inertia is greater than the minimum inertia demand.

As shown in Fig. 6, 15 virtual inertial devices were introduced into the system. When the system is disturbed, it can output active power

Table 3
Inertia under Different Degrees of Penetration.

Penetration (%)	Renewable energy (MW)	Inertia (s)	Kinetic energy (MWs)
10	1480	2.45	36,343
15	2220	2.32	34,323
20	2960	2.18	32,304
25	3700	2.05	30,285

Table 4
Upper and Lower Bounds.

penetration	Lower bound		Upper bound	
	Inertia (s)	Kinetic energy (MWs)	Inertia (s)	Kinetic energy (MWs)
10%	2.45	36,343	3.5	51,919
15%	2.32	34,323	3.3	49,033
20%	2.18	32,304	3.1	46,149
25%	2.05	30,285	2.9	43,264

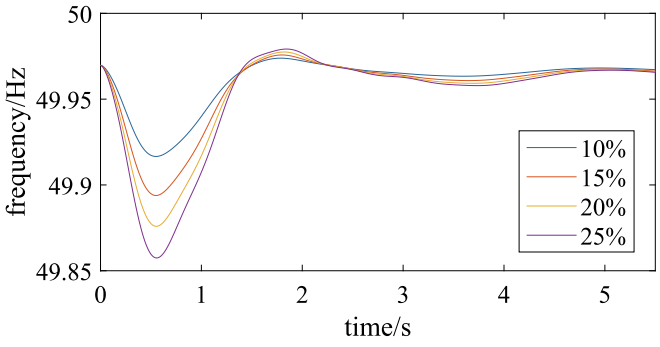


Fig. 8. System frequency of different degrees of renewable energy penetration under a disturbance of 200 MW.

Table 5
Maximum Frequency Deviation.

penetration	disturbance	$f1$	$f2$	$\triangle f$
10%	200 MW	49.97 Hz	49.92 Hz	0.05 Hz
15%		49.97 Hz	49.89 Hz	0.08 Hz
20%		49.97 Hz	49.88 Hz	0.09 Hz
25%		49.97 Hz	49.86 Hz	0.11 Hz
10%	1000 MW	49.97 Hz	49.82 Hz	0.15 Hz
15%		49.97 Hz	49.79 Hz	0.18 Hz
20%		49.97 Hz	49.75 Hz	0.22 Hz
25%		49.97 Hz	49.69 Hz	0.28 Hz
10%	2500 MW	49.97 Hz	49.65 Hz	0.32 Hz
15%		49.97 Hz	49.59 Hz	0.38 Hz
20%		49.97 Hz	49.54 Hz	0.43 Hz
25%		49.97 Hz	49.45 Hz	0.52 Hz

Table 6
Upper and Lower Bounds of the Minimum Inertia Requirement.

Lower bound		Upper bound	
Inertia (s)	Minimum inertia requirement (MWs)	Inertia (s)	Minimum inertia requirement (MWs)
1.60	23,750	2.11	31,250

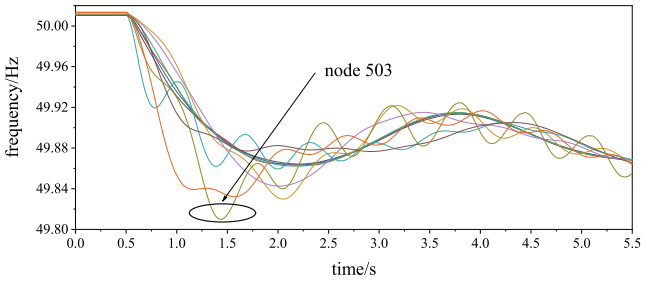


Fig. 9. Frequency response without grid-forming VI.

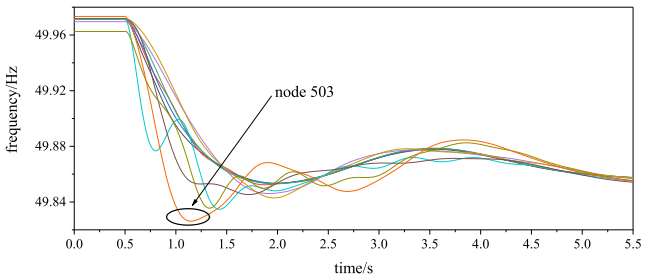


Fig. 10. Frequency response with grid-forming VI.

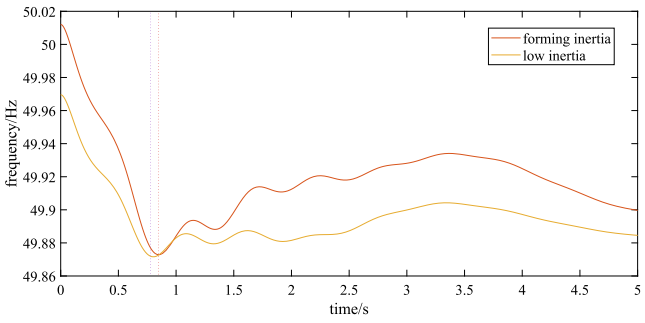


Fig. 11. Comparative analysis of the frequency responses without and with grid-forming VI.

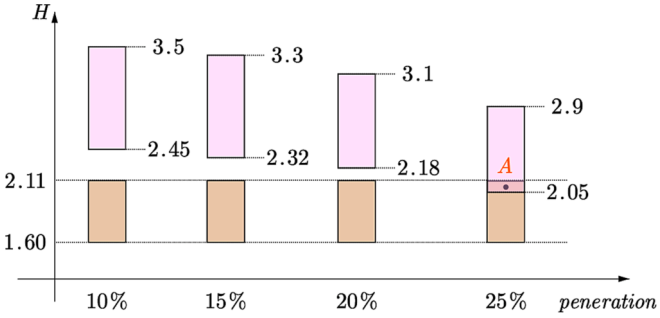


Fig. 12. Upper and lower bounds.

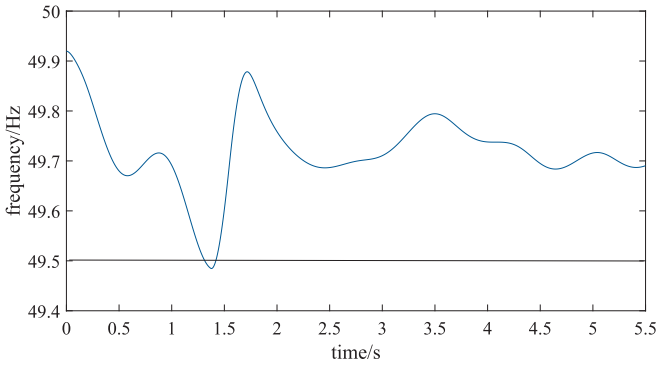


Fig. 13. Frequency response curve of point A.

without delay to stabilize the fluctuation, which is equivalent to reducing the unbalanced power to reduce the minimum inertia requirement of the power system. When $P_{VI} = 40MW$, the calculation indicates that when $\Delta P = 2500$MW and $RoCoF_{max} = 2Hz/s$, the minimum inertia of the system can be estimated as follows in Table 6

The frequency responses with and without grid-forming VI are shown in Fig. 9 and Fig. 10, respectively, according to which the frequency of each generator node decreased after the disturbance.

As shown in Fig. 9 and Fig. 10, when the system was disturbed, the frequency deviation of Node 503 was the most significant. According to the analysis in Fig. 9, this satisfies the expectation. Owing to the weak connection characteristics between Area 5 and other areas, the proportion of synchronous generators in the area was relatively small, its frequency fluctuation was significant, and the influence of the center of inertia must be considered. The analysis is based on the frequency curve of Node 503, as follows:

As shown in Fig. 11, Based on the comparison, the frequency of the system with virtual inertia reaches the lowest point more slowly and recovers better, indicating that the virtual inertia device is crucial in improving the frequency stability of the power system.

As shown in Fig. 12, when the penetration was 25%, frequency instability occurred, that is, at operation point A. The frequency–response curve is shown in the figure below.

As shown in Fig. 13, the lower bound of the predicted system inertia value is smaller than that of the minimum inertia requirement provided the actual operating point is located near A. This causes the frequency to trigger a threshold under the frequency load shed.

5. Conclusion

This study considered grid-forming virtual inertia devices equipped with a low-inertia power system that can provide FFR to support the frequency of the power system. First, this study proposed a power system inertia estimation method that considered renewable energy penetration and provided the upper and lower bounds of system inertia, which can be used to determine the frequency stability. After the power system is disturbed, the minimum inertia requirement was obtained based on the *RoCoF*. The upper and lower bounds of the requirement were provided, which was convenient for dispatchers to control urgently. Finally, data analysis and simulation validation of the model were conducted and the main conclusions are as follows:

(1) The inertia levels of power systems with different degrees of penetration differed and unit commitment influenced the prediction of inertia, which could be conveniently expressed in intervals. Based on the simulation results, the lower bound of inertia estimation was crucial.
(2) Grid-forming virtual inertia could provide the necessary power support, which is conducive to system frequency stability. VI devices could significantly increase the system inertia level and their FFR can reduce the minimum inertia requirement.
(3) In large-scale power systems, there are numerous synchronous machines with different inertia constants, and the proportion of new energy is uncertain, resulting in the inability to find the exact center frequency of inertia. Therefore, fully utilizing the fast response capability of converter resources, combined with the residual inertia response capability of traditional units, can assist dispatchers in monitoring the safety of the power system.

The center of inertia and coordination between the virtual inertia equipment and conventional units should be further investigated.

CRediT authorship contribution statement

Guihong Zhang: Writing – original draft. **Junzhi Ren:** Writing – original draft. **Yuan Zeng:** Writing – review & editing. **Fei Liu:** Resources. **Shibin Wang:** Data curation. **Hongjie Jia:** .

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

The data that has been used is confidential.

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